Marcos **Benevides** | Software Engineer

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Executive Summary

Software engineer with over 8 years of experience focused on **functional programming** and **declarative infrastructure**. My work seeks to converge code, academic concepts and operations through solid mathematical foundations, leveraging the most robust aspects of Computer Science to create resilient systems.
I'm here to deliver only what is promised, with simple abstractions and minimal overhead.

Employment

TradeKart — **London, United Kingdom (Remote)**
Software Engineer, Contractor — *Jan 2026 – Present*
- F# Backend Developer and DevOps.
- Helped CTO and new team stabilize multiple services.

Terrateam — **Amsterdam, North-Holland (Remote)**
OCaml Developer, Contractor — *Apr 2025 – Dec 2025*
- Originally joined as a freelancer to implement SCRAM authentication for PostgreSQL (to improve secure database connections in their custom PostgreSQL library).
- Refined comment-related features that drive workflows for the main TACOS (Terraform Automation and Collaboration Software) product.

Divisions Maintenance Group — **Ohio, United States (Remote)**
Technical Lead, Full-Time Employee — *Jul 2024 – Apr 2025*
- I was promoted to tech lead of one of our financial services.

Divisions Maintenance Group — **Ohio, United States (Remote)**
Software Development Engineer 2, Full-Time Employee — *Apr 2022 – Jul 2024*
- F# and C# Backend Developer.
- Exorcised demonic bugs and improved a legacy monorepo by troubleshooting issues in a decade-old database.
- Implemented new projects to transition from our outdated systems, incorporating immutable architectures based on functional programming principles.
- Created specialized tools to facilitate local issue reproducibility within our microservices infrastructure.

datarisk.io — **Florianópolis, Brazil (Remote)**
Lead DevOps Engineer, Contractor — *Nov 2020 – May 2022*
- Led an effort to move cloud providers, inherited a poorly maintained, scarcely monitored and undocumented AWS infrastructure. In less than 2 years we had moved to Azure, with proper monitoring, cost-saving measures and an infra-as-code approach.
- Reduced our cloud-billing by 80%.
- Mentored my interns until they were able to safely replace me (and be promoted).
- F# and Python Backend Developer.

datarisk.io — **Florianópolis, Brazil (Remote)**
Software Engineer, Contractor — *Oct 2019 – Nov 2020*
- Started as the 3rd engineer (and first remote hire), focusing on Python Backend development before being promoted to Lead DevOps Engineer.

FIEMA/CEUMA — **São Luís, Brazil**
Internships and Java Development Gig — *Feb 2018 – Nov 2018*

Skills

Architecture Design Monolithic and Distributed solutions.
Cloud AWS, Azure and Oracle.
DBA *Languages* (SQL, Prolog), RDBMS (PostgreSQL, SQL Server, SQLite)
Devops *Infra* (Linux, NixOS), *Tools* (Docker, Git, Kafka, Kubernetes, Makefile, Shell, Terraform), Automation (CI/CD).
Management Technical Leadership and Mentoring.
Programming *Functional* (Erlang, Gleam, F#, Haskell, Nix, OCaml), *Object Oriented* (C#, Python), *Procedural* (C, C++).

Education

UFMA - Universidade Federal do Maranhão **São Luís, Brazil**
B.S. in Computer Science *Apr 2013 – Jun 2019*

During my undergraduate years, I didn't limit myself to what was in my official curriculum. I explored wide range of topics, was an active member of a Study Group on Logic and Formal Philosophy, a multi-disciplinary initiative headed by our Department of Philosophy. I've also received two scholarships to study the following subjects:

- Introductory Real Analysis & Differential Geometry within our Department of Mathematics (2013-2014).
- Computer Vision & Image Processing as a undergraduate in Computer Science (2014-2016).

Projects

- I contribute sporadically and maintain some packages on NixOS.
- Core developer of Lyceum, an experimental MMO RPG built atop Erlang, Zig, Nix/NixOS and PostgreSQL (with omnigres). Also maintain Chessboard, the NixOS-based server where we run our game's live demos.
- I also develop and maintain some Erlang/Gleam-based packages on hex, like: eparch, erlandono and migraterl.

Talks

- I did a talk about Erlang, PostgreSQL and NixOS on Gambiconf 2025.